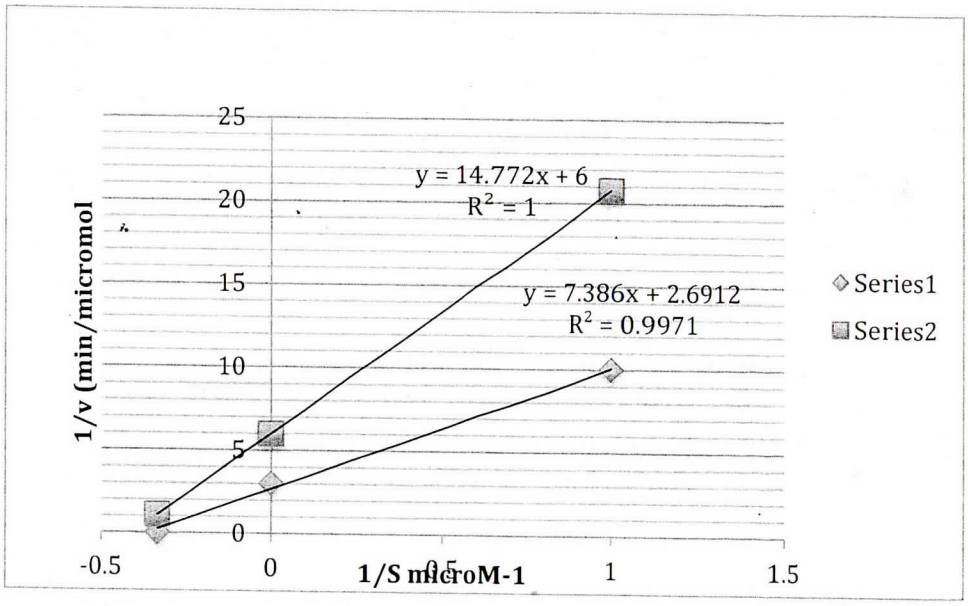
Biochemistry Cume, March 8, 2019

[25pts each] For the next 4 questions, consider the plot drawn. A Lineweaver-Burke plot is constructed for the enzyme (diamond; series 1) and the enzyme with inhibitor (square; series 2).



1. What is the V_{max} (μmole/min) and V_{max}^{app}?

2. What is the K_M (μM) and K_M^{app}?

3. The enzyme contains one catalytic subunit that consists of 200 amino acids. Each solution contained 0.1ng. What is the turnover number (second^{-1}) without inhibitor? Weighted average molecular weight per amino acid is 110 Dalton.

4. What is the K_i (μM) of the inhibitor when [I] was 0.1 μM?

Type of inhibition	V_{max}^{app}	V_{max}^{app}/K_M^{app}	K_M^{app}
competitive	V_{max}	$\dfrac{V_{max}/K_M}{\left(1+\frac{[I]}{K_i}\right)}$	$K_M\left(1+\frac{[I]}{K_i}\right)$
mixed	$V_{max}/(1+\frac{[I]}{K_{iu}})$	$\dfrac{V_{max}/K_M}{\left(1+\frac{[I]}{K_i}\right)}$	$K_M\left\{\frac{1+([I]/K_i)}{1+([I]/K_{iu})}\right\}$
noncompetitive	$V_{max}/(1+\frac{[I]}{K_i})$	$\dfrac{V_{max}/K_M}{\left(1+\frac{[I]}{K_i}\right)}$	K_M
uncompetitive	$V_{max}/(1+\frac{[I]}{K_{iu}})$	V_{max}/K_M	$K_M/(1+\frac{[I]}{K_{iu}})$